

Cheap Signals, Costly Proof: The Reach and Limits of Award-Layer Screening in Cartel Enforcement

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Abstract

Cartel enforcement must allocate costly proof-producing effort before legal proof exists, drawing first on administrative records that are cheap but legally thin. How far can such records screen? Using São Paulo’s BEC procurement platform (2009–2019) and CADE adjudications, we build a minimal award-layer construct—persistent zero-win participation among always-loser firms, the *frequent losers*—and subject it to a decomposition that most screening studies omit. The construct concentrates adjudication-anchored exposure to adjudicated cartel environments; the decomposition then draws a sharp boundary. Much of its apparent power reflects procurement opportunity and a single adjudicated case, and it orders firms retrospectively among incumbents rather than prospectively across the platform. Our contribution is a transferable method for separating genuine screening signal from exposure arithmetic and case concentration, and a map of where cheap administrative records can, and cannot, order forensic priority. Liability remains in the richer bid-level record.

Keywords: cartel screening, public procurement, award-layer data, enforcement triage, incomplete observability, bid rigging

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1. Introduction

Cartel enforcement begins before legal proof exists. An agency deciding where to investigate rarely starts with the communications, bid histories, leniency testimony, or tender-level conduct evidence that would support liability. It starts with thinner information: administrative records, complaints, procurement anomalies, repeated interactions, and case leads that make some firms and markets worth a closer look. The first legal-economic problem is therefore not proof itself. It is the allocation of proof-producing effort.

That distinction matters in procurement. The empirical cartel-screening literature usually starts from the bid layer: submitted bid amounts, ranks, dispersion, histories, and sometimes bidder networks. Classic screens compare bidding under competition and collusion ([Porter and Zona, 1993](#); [Bajari and Ye, 2003](#)); variance and descriptive screens formalize suspicious bid distributions ([Abrantes-Metz et al., 2006](#); [Harrington, 2008](#)) and field-test them in procurement ([Imhof et al., 2018](#); [Imhof, 2019](#)); recent work extends the approach through bidder groups ([Conley and Decarolis, 2016](#)), machine learning ([Huber and Imhof, 2019](#); [Wallimann et al., 2023](#)), and robust screen design ([Chassang et al., 2022](#)). Those objects are close to the evidentiary stage but not always cheap at scale: many procurement systems make contract-award records routinely visible while keeping per-bidder bid files in operational systems that require case-specific recovery. An enforcement agency then faces a sequencing problem—before opening the costly bid layer, can it use the cheap award layer to decide where bid-layer forensics should begin?

This paper studies that question on São Paulo’s BEC procurement platform, covering 1,654,401 tender-items from 2009 to 2019. BEC records a routine award layer—participants, winners, item codes, negotiated prices, procuring units, years, and modalities—and also preserves a richer bid layer recoverable through the LANCES export. CADE procurement-cartel adjudications provide external legal anchors. This combination lets us study an earlier stage than most bid-distribution screens: the point at which an agency observes award records but has not yet recovered the bid-level evidence needed for forensic evaluation.

The award-layer statistic is deliberately simple. Among firms with zero wins, we rank participation intensity using $\log(1 + \text{tenders_count})$ and implement an auditable binary rule that flags *frequent losers*. The logic is monotone rather than classificatory: if coordinated bidders use loser-side participation to create the appearance of competition while avoiding the win, persistent zero-win participation can rank loser-side exposure under a transparent behavioral condition. The statistic does not identify a firm type or assign cartel membership; the rank allocates forensic attention, while the richer bid layer remains where tender-level conduct and agreement are evaluated.

That legal scope determines the validation target. Direct CADE defendants are legal membership anchors, but they are not the natural object of a zero-win loser-side screen. Procurement cartels often allocate winning roles, rotate designated winners, and use other firms on the losing side of the tender (Pesendorfer, 2000; Asker, 2010); the same role-allocation logic is central in cartel theory and case evidence (Marshall and Marx, 2012), and large-scale procurement evidence shows how such arrangements appear in administrative bidding records (Kawai and Nakabayashi, 2022). We therefore validate the screen against always-loser firms that bid alongside direct CADE defendants in adjudicated cartel environments. These cobidders define an adjudication-anchored exposure target: direct defendants anchor legally adjudicated environments, and cobidders are the loser-side footprint the award layer can plausibly rank before bid-level proof is recovered.

The empirical design is organized around the threats that would make such a screen uninformative. A skeptic’s first objection is mechanical: high-volume zero-win firms may simply have more opportunities to meet CADE defendants, so a raw co-bidding association could reflect participation volume or opportunity-set exposure rather than signal. We discipline the ranking against participation volume, opportunity exposure, timing/leakage, case composition, and a bid-layer benchmark; Section 4 and the appendices report the battery, and Appendix C gives the exact exposure, leakage, and timing audit construction.

The results support a limited but operationally useful claim. The frequent-loser ranking concentrates adjudication-anchored cobidders inside the always-loser stratum, including when the score is formed before the target window. Much of this raw con-

centration reflects procurement opportunity—firms that participate more, in the same environments as adjudicated defendants, are mechanically more exposed to them—but a limited and statistically significant signal survives once opportunity is held fixed. It performs poorly against direct CADE defendants, consistent with its loser-side scope rather than a generic cartel-membership interpretation. Cobidders inside the stratum also look different from other frequent losers: deployed more broadly, closer to legal cartel anchors, in more concentrated product portfolios, with bid-level patterns consistent with, but not diagnostic of, credible losing roles—giving the exposure target economic content beyond ordinary high-volume losing.

A complementary result concerns the division of labor between the two layers. On the same adjudication-anchored target, the award-layer score carries information beyond an Imhof–Wallimann-style bid-distribution benchmark, so the layers enter at different stages: award records prioritize where to look, and bid records evaluate what is found. Used as a gatekeeper, the award-layer ranking traces a cost-recall frontier: across a range of survivor-pool sizes it preserves much of the adjudication-anchored recall of full bid-layer scoring while reducing the bid-recovery footprint. The reduction is real but honest about its denominator—large in the count of firms opened, far smaller in tender-items and bid rows, because the survivors are high-participation firms. Joint scoring is a full-observability upper bound; the frontier, not a universal cutoff, is the enforcement-design object.

This paper makes three contributions. First, it reframes procurement-cartel screening as evidence allocation under costly observability: existing bid-distribution screens evaluate suspicious bidding once bid-level data have been recovered, while we study the prior institutional stage, where agencies observe award records but must decide where recovery should start—turning screening toward enforcement triage, not toward treating a statistical flag as proof (Harrington, 2008; Sánchez Graells, 2019; Chassang et al., 2022). Second, and methodologically, we subject a minimal award-layer construct to a decomposition screening studies usually omit: not whether it attains a high pooled AUC, but how much of that performance is genuine ranking signal rather than mechanical opportunity exposure, retrospective information, or concentration in a single adjudicated

case—pairing pre-committed rejection regions with opportunity-cell adjustment, strict rolling-origin timing, and leave-one-case-out tests. Third, that decomposition yields a map of reach and limits: persistent zero-win participation does concentrate a legally limited, adjudication-anchored loser-side exposure target, but its discriminating power is largely retrospective and incumbent-specific, with a substantial share attributable to procurement opportunity and to one case. Cheap administrative records can order forensic priority *within* these boundaries—and locating the boundaries is itself the enforcement-design product.

The boundaries are part of the design. The award layer ranks forensic priority; the validation target is adjudication-anchored exposure; and the bid layer remains the proof-producing stage. Price evidence is scope information, not a damages exercise. Strategic response also shapes implementation: a static cutoff invites gaming, while refreshed thresholds, continuous ranks or capacity-constrained queues, bunching checks, and bid-layer follow-up fit the staged enforcement architecture.

The broader lesson is institutional. A legal system that waits for full proof before ranking investigative priorities will spend scarce forensic capacity poorly; a system that treats cheap suspicion as proof will overreach. The useful middle ground is a disciplined triage rule—cheap enough to run before proof exists, informative enough to order forensic attention within a known reach, and legally limited enough to leave liability where it belongs, in the richer evidentiary record. Knowing where such a rule stops ranking is as much a part of disciplined enforcement as knowing where it begins.

2. Costly Proof and Observable Awards

The paper’s empirical object is an institutional sequence, not a stand-alone classifier. An enforcement agency first observes a thin administrative record, then decides whether to spend resources recovering richer bid-level evidence, and only later can that evidence contribute to legal proof. This section defines that sequence for BEC procurement: the legal role of a screen, the award and bid information layers, and the adjudication-anchored labels used to validate a loser-side ranking.

2.1. Screening Before Proof

Antitrust agencies do not choose investigation under full information. They choose it while the facts needed for liability are still costly to obtain. In a procurement-cartel case, proof requires evidence of coordinated conduct: communications, allocation rules, bid rotation, cover bidding, or tender-level patterns inconsistent with independent rivalry. Brazilian competition law treats bid rigging as an infraction under Lei 12.529/2011; comparable frameworks likewise require evidence beyond suspicious administrative patterns (Hovenkamp, 2005). A pattern in award records does not contain those legal predicates.

The institutional question is therefore earlier than proof and narrower than liability. The Becker–Stigler tradition treats enforcement intensity, investigation, and sanctions as costly instruments (Becker, 1968; Stigler, 1970), and cartel-screening work makes the same allocation logic operational: when investigation is costly and misconduct is only partially observable, agencies should use cheap screens to decide where expensive inquiry has the highest expected return (Baker, 2003; Harrington, 2008; OECD, 2022); leniency policy illustrates the same trade-off (Wils, 2007, 2016). For procurement cartels, the first-stage problem is not whether a cheap statistic can decide liability, but whether it can order the queue for recovering proof-producing evidence.

This is the sense in which the paper studies triage rather than classification. A model that uses submitted bid amounts, bid ranks, and within-tender bid moments assumes the forensic layer has already been opened—the right information set for evaluating suspicious bidding, but not for deciding where the first costly bid-file request should be sent. An award-layer screen has value only at this earlier point: its output is *forensic priority*, not an accusation. A useful triage statistic must be computable from records available before case-specific bid recovery; must rank firms or environments where richer evidence of cartel conduct is more likely to be worth recovering; and must remain legally limited, so that administrative suspicion does not substitute for proof. The BEC setting supplies the two data layers and the external adjudication anchors needed to evaluate whether an award-layer statistic can play that role.

2.2. Award and Bid Layers in BEC

São Paulo’s BEC platform preserves the two information layers an enforcement sequence requires. The sample contains 1,654,401 tender-items from 2009 to 2019, dominated by two modalities: *Convite*, a sealed-bid invitation procedure under Lei 8.666/1993 with a statutory three-bidder quorum and a low value cap; and *Pregão*, a reverse electronic auction under Lei 10.520/2002 with real-time offers and no minimum-bidder rule.² Both generate a routine award record before any case-specific bid recovery occurs. Auction format, bureaucratic incentives, discretion, and rules can shape both bidding behavior and observed outcomes (Athey et al., 2011; Decarolis, 2014; Coviello and Gagliarducci, 2017; Decarolis et al., 2020, 2025).

The *award layer* is that routine administrative envelope. It records participant identity, winner identity, item code, negotiated price, procuring unit, year, and modality. These fields show who appeared, who won, what was bought, and where a firm repeatedly participated without winning. They do not reveal the bids submitted by each participant, the timing of offers, bid revisions, or within-tender dispersion. The award layer is therefore cheap enough for broad monitoring, but too thin to supply legal proof of coordination.

The *bid layer* is the forensic-recoverable layer. BEC preserves per-bidder offers, timestamps, and bid-revision sequences in the LANCES export. Those records support bid-distribution forensics: within-tender dispersion, skewness, spread, distance from the winning bid, and related Imhof–Wallimann-style features (Imhof et al., 2018; Imhof, 2019), including machine-learning variants that combine these bid moments at scale (Huber and Imhof, 2019; Wallimann et al., 2023). But these features require the bid file to be recovered, cleaned, linked, and inspected; they are not available at the same cost or timing as an award-record query. The empirical design treats that cost difference as substantive: the bid layer is available but costly—feasible for case work and validation, not the natural first object for system-wide triage. The issue is cost, timing, and processing burden, not data non-existence.

²Throughout, a firm’s participation is counted once per tender-item it enters. In *Pregão*, a bidder may revise its offer many times within a single tender-item; these iterative offers do not increase the participation count T_i , which is defined at the tender-item level.

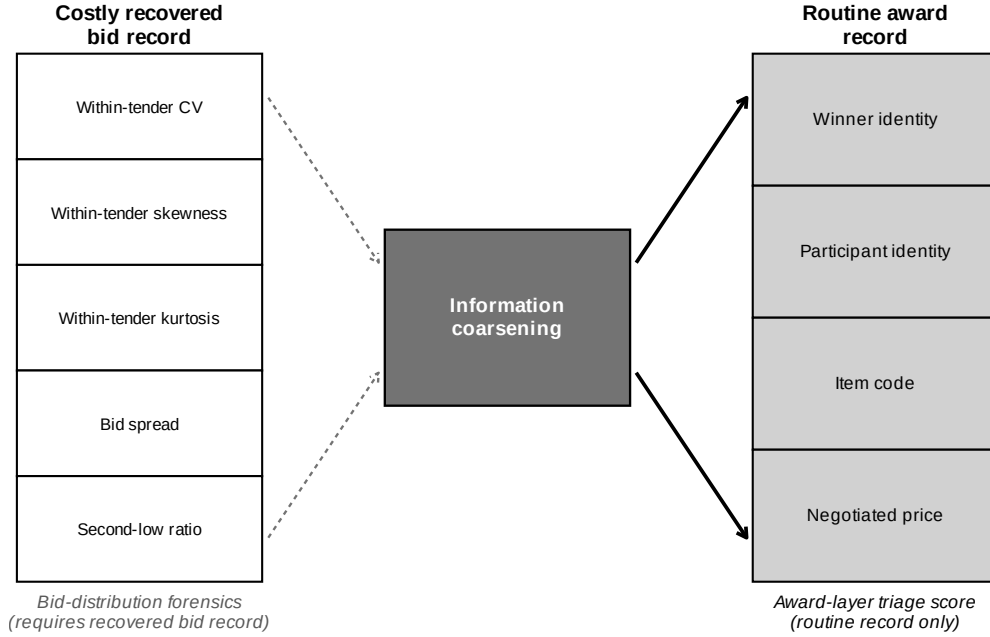


Figure 1: Information layers in procurement-cartel enforcement. This is an information-cost diagram, not a detector horse race. Bid-distribution forensics requires per-bidder bid amounts and within-tender bid moments; the award-layer screen uses participant identity, winner identity, item code, and negotiated price. The bid layer is not absent: it is costly to recover, clean, and inspect from the LANCES export. The question is whether the cheap award layer can rank where the costly bid layer should be opened first. A pattern in award records does not imply legal proof of coordination.

Alt text: Two vertical columns of stacked boxes represent two information layers. The left column lists costly recovered bid-record features (within-tender bid moments such as dispersion, skewness, spread, and second-low ratio). The right column lists routine award-record fields (winner identity, participant identity, item code, negotiated price). A central gate links them: the thinner award layer is available earlier and more cheaply and is used to rank where the costly bid layer should be recovered first.

Figure 1 summarizes the distinction: an information-cost diagram, not a detector horse race. Bid-distribution methods remain the natural forensic tools once bid microdata have been recovered; award-layer triage asks whether the thinner routine record can rank where that recovery should start.

2.3. CADE Anchors and Adjudication-Anchored Exposure Labels

CADE adjudications supply the legal anchors, not the screen’s labels mechanically. The linked BEC portfolio contains 12 adjudicated procurement-cartel cases with 65 firm-defendants, of which 47 are active in BEC during the sample window. Final decisions span 2015–2025, with long process-to-ruling lags. That timing is central: if a cheap screen is useful, it is because it can move investigative attention before the completed adjudication record exists.

The adjudication record creates two objects with different roles. Direct CADE defendants are legal membership anchors: firms named in adjudicated procurement-cartel cases and matched to BEC participant records. They locate environments where legal process established cartel conduct, but they are not the natural target of a zero-win loser-side statistic—defendants may occupy winner-side, allocation, or rotating roles, and a screen built from repeated losing should not be judged as if it were a generic cartel-membership detector.

The validation target is narrower. A cobidder is an always-loser firm that appears in at least one BEC tender-item with a direct CADE defendant, yielding 193 always-loser cobidders. Cobidder status is not legal membership or a latent truth about agreement. It is an adjudication-anchored exposure label: the firm is observed on the losing side near a legally established cartel anchor, appropriate for evaluating forensic priority, not liability. Collapsing defendants and cobidders into a single cartel class would overstate what award records establish; requiring a loser-side screen to recover direct defendants would misstate what the statistic is designed to rank. The relevant question is whether persistent zero-win participation concentrates the loser-side firms most plausibly worth bid-layer follow-up around adjudicated cartel environments. Because adjacency can also arise from opportunity—firms active in the same items, years, modalities, or buyers as

CADE anchors have more chances to meet them—the validation design later disciplines participation volume and opportunity-set exposure separately from the ranking signal.

Table 1 fixes the roles used throughout the paper: the outer BEC population, the always-loser stratum the award layer can rank, the frequent-loser operational screen, the direct-defendant legal anchors, and the cobidder loser-side validation target.

Table 1: Data Layers, Populations, and Legal-Economic Roles

Object	Size / layer	Role in the analysis	Legal interpretation
Award layer	Routine record	Participants, winners, item codes, prices, procuring units, years, and modalities; the cheap triage layer.	Administrative record; not proof.
Bid layer	LANCES export	Per-bidder offers, timestamps, revisions, and bid moments; the costly forensic layer.	Forensic record; can support bid-level analysis.
All BEC participants	41,444 firms	Outer firm universe for award-record construction.	Outer universe.
Always-losers	16,843 firms	Zero-win stratum in which the award-layer screen ranks loser-side exposure.	Screening stratum.
Frequent losers	2,735 firms	Operational first-stage flag within always-losers; not a legal category.	Operational flag; not a legal category.
Direct CADE defendants in BEC	47 firms	Legal anchors from adjudicated procurement-cartel cases.	Legal anchor.
Always-loser cobidders	193 firms	Firms losing alongside direct defendants; validation target.	Exposure label; not membership.

Notes: Direct CADE defendants are legal anchors. Always-loser cobidders are adjudication-anchored exposure labels. Frequent losers are an operational screen, not a legal category. The categories separate legal membership from screening exposure. Benchmark-specific cobidder counts and the funnel reconciliation appear in Table 2.

The rest of the paper keeps these objects separate: the screen operates inside the always-loser stratum, the validation target is cobidder adjacency, direct defendants anchor the environments without defining the screen’s success, and bid-layer forensics remains the proof-producing stage. This separation lets the paper evaluate a legally limited triage architecture without turning suspicion into adjudication.

3. Award-Layer Triage

The award layer observes outcomes before it observes conduct. A losing bid may be sincere, poorly priced, capacity constrained, exploratory, strategic, or coordinated. The screen therefore uses what the award layer can measure at the triage stage: which firms appeared, which won, and which kept appearing without ever winning. The object is a ranking for forensic priority, not proof of conduct or a classification of cartel membership. This section defines that ranking, the operational queue it yields, and the implications that discipline it.

3.1. Persistent Zero-Win Participation

Let T_i denote the number of BEC tender-items in which firm i participates, and let W_i denote the number it wins. In Pregão, a firm may revise its offer repeatedly within one tender-item; T_i counts tender-item participations, so these iterative offers do not raise T_i . The award-layer screen operates inside the zero-win stratum, $\{i : W_i = 0\}$, which contains 16,843 firms. Within that stratum, the continuous score is

$$s_i = \log(1 + T_i).$$

The log transformation stabilizes the scale but does not change the ordering. The economic object is the rank: among firms the award layer observes only as losers, which ones appear persistently enough to warrant bid-layer follow-up?

The maintained condition is monotonic rather than classificatory. If cartel-deployed losing participants are present, they should appear more persistently in the zero-win stratum than ordinary losing firms, because such a role is useful only through repeated deployment in cartel-targeted tenders. The condition is maintained, not derived; Appendix [Appendix B](#) formalizes it as a monotone-likelihood-ratio ranking result. The main text uses it more modestly: persistent zero-win participation is a cheap statistic that can rank loser-side exposure under a transparent behavioral assumption and direct attention to the bid layer where conduct can be evaluated. It does not assign a firm type or cartel membership.

The zero-win restriction is a scope restriction: many firms never win for ordinary competitive reasons (poor fit, high costs, capacity constraints, market exploration, subcontracting positioning, or bad luck), and the screen’s content comes from persistence inside that heterogeneous loser-side set. Appendix [Appendix B](#) adds the dynamic margin behind this: ordinary zero-win firms face participation costs and may exit after repeated losses, while a losing role that carries strategic value or compensation can sustain persistence, and a platform-survival audit finds that frequent-loser cobidders remain active markedly longer than other frequent losers—consistent with persistence as a triage signal, not a legal category. If the ranking has enforcement value, it should concentrate adjudication-anchored loser-side adjacency beyond what generic participation volume, product-market overlap, and opportunity to meet CADE anchors would mechanically produce.

3.2. From Ranking to an Operational Queue

The continuous score is the central object because agencies can translate a rank into different investigative protocols: inspect the top k firms under fixed capacity, refresh the rank each quarter and examine bunching below old cutoffs, or request LANCES bid files for a survivor pool before applying bid-distribution forensics. All preserve the same sequencing logic: award records first, costly bid recovery second.

For an auditable binary implementation, we also report a frequent-loser flag marking zero-win firms whose participation count exceeds the median plus 1.5 times the interquartile range of the always-loser participation distribution. In BEC, the statistical cutoff is 13.5, implemented as $T_i \geq 14$, which flags 2,735 firms, or 16.2% of the always-loser stratum. The cutoff is fixed from the participation distribution alone, before any CADE label enters the analysis.

The $T_i \geq 14$ rule is an auditable administrative implementation of the rank, not a structural or legal threshold. The paper does not rest on the number 14 but on the continuous loser-side participation ranking. Two analysts applying the same rule to the same award records obtain the same first-stage list; above the cutoff does not mean cartel member, and below it does not mean clean. Because a published static threshold can be gamed, the enforcement architecture treats the binary flag as one implementation among

several—continuous ranks, top- k queues, refreshed thresholds, and budget-calibrated survivor pools—and the strategic-adaptation analysis later returns to this point.

3.3. Testable Implications

The screen is useful only if it survives the threats implied by its limited information. The core implication is the validation claim: persistent zero-win participation should rank adjudication-anchored loser-side cobidders within the always-loser stratum, beyond what mechanical exposure—high participation volume and operation in the same procurement environments as CADE defendants—would produce under ordinary competition. The remaining implications are scope (the statistic should perform worse against direct defendants than cobidders, ranking a different role), economic content of the cobidder target, non-redundancy of the award layer relative to bid-distribution forensics, gatekeeping value at affordable budgets, and vulnerability of the published binary threshold to strategic adaptation. These testable implications, and the threats that would weaken each, are evaluated in Section 4 and the appendix; the screen is assessed at the stage for which it is designed—the allocation of costly forensic attention before liability evidence is assembled. The BEC cutoff is an implementation detail; the portable object is the staged architecture that uses cheap administrative records to discipline where costly cartel forensics should begin.

4. Validating the Loser-Side Ranking

This section validates the award-layer ranking, not the legal status of ranked firms. The validation target is the one defined in Section 2.3: always-loser firms observed alongside direct CADE defendants in adjudicated procurement-cartel environments. These cobidders are adjudication-anchored exposure labels, not legal members of a cartel. A useful ranking must do more than report a high AUC. The main threat is exposure—firms that participate often, or operate in the same environments as defendants, have more opportunities to appear near those anchors even under ordinary competition, so a screen must concentrate loser-side firms beyond what participation volume and opportunity alone imply (Porter and Zona, 1993; Bajari and Ye, 2003; Conley and Decarolis, 2016;

Kawai and Nakabayashi, 2022). Two further threats are timing (the screen should rank before the target information is used) and legal overreach (it should not be read as a generic cartel-membership detector).

Subsection 4.1 fixes the label-construction funnel and reconciles the validation samples. Subsection 4.2 is the central exercise: it disciplines the ranking against procurement opportunity and isolates the score’s residual contribution net of exposure. Subsection 4.3 imposes strict timing and case-composition discipline, with the sobering result that the operational ranking is a retrospective, incumbent-pool, largely single-case diagnostic. Subsection 4.4 folds the direct-defendant scope check into a bounded reading of what the validation establishes.

4.1. Label Construction and Sample Reconciliation

The full CADE legal portfolio comprises 12 adjudicated procurement-cartel cases with 65 legal firm-defendants. We match defendants to BEC participant records by exact 14-digit CNPJ equality, with no fuzzy or manual step; 47 are active in BEC and define the direct-defendant set. Their defendant tender-items (52013 distinct items) locate the cartel-affected environments, within which a cobidder is a zero-win firm sharing at least one tender-item with a direct defendant.

The main validation target is the 193 always-loser frequent-loser cobidders defined under a narrow cartel-tender restriction, which we keep as the primary benchmark for comparability with prior drafts. We disclose one replication caveat cleanly: that target’s original builder is archived and is not reproduced by current scripts. A fully scripted alternative —counting any always-loser sharing a tender-item with a BEC-active direct defendant over the full portfolio—yields 341 frequent-loser and 651 always-loser cobidders, overlapping the main target in 149 firms; we carry it as the reproducible robustness funnel. The conservative pre-2020 benchmark is not a mechanical subset of the main target: it restricts the portfolio to the 4 early cases but applies the broad cobidder definition over the wider always-loser stratum, yielding 208 cobidders (107 frequent losers) from 19 defendants. Under a common definition the subset relation is restored. Table 2 reports every benchmark side by side.

Table 2: Label-Construction Funnel and Benchmark Reconciliation

Sample / benchmark	Cases	Legal def.	BEC-active def.	AL cobid.	FL cobid.	Cobidder def.	Reason vs main target
Full CADE portfolio	12	65	47	651	341	broad	reference
Main target — static narrow	12	65	—	—	193	narrow	primary
Transparent broad funnel	12	65	41	651	341	broad	robustness
Conservative pre-2020	4	—	19	208	107	broad	early-cases subset

Notes: A cobidder is an adjudication-anchored exposure label, not membership. The main target (193, narrow cartel-tender definition) has an archived builder; the broad shared-tender-item funnel yields 341 frequent-loser and 651 always-loser cobidders. The conservative benchmark reproduces to 208/107. The 47 cited active defendants correspond to 48 distinct crossmatch CNPJ, of which 41 are firm_tender_map-active. Counts are unique firms.

Before adding stricter discipline, a baseline question: do firms flagged by the award-layer rule appear disproportionately among always-loser firms near adjudicated anchors? They do. In the conservative pre-2020 benchmark, the observed frequent-loser share among cobidders is 3.95% against a participation-stratified placebo mean of 1.24% (excess ratio 3.2 $\times$, $p < 0.001$), with ROC-AUC 0.748 (95% DeLong CI [0.713, 0.783]) (DeLong et al., 1988). Persistent zero-win participation is not random with respect to adjudication-anchored loser-side exposure. But a high AUC here is not yet enough, because participation volume and shared environments mechanically manufacture proximity. The next subsection therefore asks how much of the ranking survives once opportunity is held comparable.

4.2. Opportunity-Adjusted Validation

This is the central validation exercise. The threat is mechanical: a firm with high participation T_i co-appears with defendants more often by construction, and a firm in the same item, buyer, year, and modality cells is more exposed even under ordinary competition. We measure opportunity directly. For each always-loser firm we count observed contact with direct defendants, O_i , and construct expected contact under random matching, $E_i = \sum_g n_{ig} p_g$, where n_{ig} is participation in cell g and p_g its defendant-density. We build E_i at three cell granularities and restrict to common support; the exposed subsample retains $n = 6,040$ always-loser firms containing all 191 positives (construction in Appendix C.1). Holding E_i fixed, we ask whether the score reproduces the label better than exposure alone, whether it adds anything net of exposure, and whether it ranks within equal-opportunity strata.

The honest answer has two parts. First, exposure is a strong baseline. An exposure-only model—no score—reproduces the label at ROC-AUC 0.946 in the exposed subsample, *above* the raw score’s 0.939, and its PR-AUC (0.194) also exceeds the raw score’s (0.126). Combined exposure-plus-score logits reach ROC-AUC near 0.98 (Table 3), but those numbers are mechanically inflated because E_i is built from the cells where defendants live and so encodes the label directly; we report them only to make the inflation visible.

Second, conditional on opportunity the score retains limited but significant residual value. A nested model that adds the score to the exposure-only model raises AUC by 0.042 (DeLong $p = 2.1 \times 10^{-6}$)—the score’s isolated contribution, not the inflated combined fit. And *within* opportunity strata, where firms are compared only against others with comparable expected contact, the continuous score still ranks the label at AUC 0.771 (FL14 0.771): well above chance and far below the raw 0.939. The score also ranks positive excess contact $O_i - E_i > 0$ at AUC 0.764, and a coarsened-exact opportunity match yields AUC 0.863 with a within-stratum gap of 0.083 in cobidder probability between flagged and non-flagged always-losers. Figure 2 shows the positive $O_i - E_i$ gap in the top score bins—the within-opportunity signal made visible.

Two exposure-preserving permutation tests close the case, in opposite directions. Under a pure-exposure binomial null the observed PR-AUC 0.126 does *not* beat the null (empirical $p = 1.00$): a pure-exposure model out-predicts the raw score on the rare-target metric. Under a permutation that shuffles cobidder labels only within matched exposure-by-participation strata, the observed PR-AUC *exceeds* the null ($p = 0.023$): once opportunity is matched, the labels are non-randomly score-ranked, and the frequent-loser enrichment beats both nulls ($p < 0.001$). Both permutation distributions are reported in Appendix [Appendix C](#).

The conservative reading is the one we carry forward. The award-layer score does not reproduce the adjudication-anchored label any better than firms’ procurement opportunity does. What it adds, net of opportunity, is a small and statistically reliable residual ordering—triage value, a reason to look first at one loser-side firm rather than another at the bid-layer stage. It is not proof of collusion and not a causal claim. The remaining threats—timing, leakage, and case composition—are taken up next: a strict-timing

battery, a leakage audit holding cobidders out of score construction (out-of-fold AUC 0.891 against an in-sample reference 0.995), and a leave-one-case-out check motivated by one composition fact: the single largest CADE case supplies roughly 47% of true positives at the top 500 of the ranking.

Table 3: Opportunity-Adjusted Validation of the Award-Layer Ranking

Design	Score	N	Pos.	ROC-AUC	PR-AUC	Prec@500	Rec@500	Lift@500	95% CI / p
Raw score	log_tc	16,843	191	0.939	0.126	0.130	0.340	11.46	[0.931, 0.946]
Raw FL14	fl14	16,843	191	0.924	0.085	0.088	0.230	7.76	[0.921, 0.926]
Exposure-only benchmark	log E_i	6,040	191	0.946	0.194	0.182	0.476	4.05	[0.818, 0.875]
Within-opportunity stratum	log_tc opp	6,040	191	0.771	—	—	—	—	pooled C
Nested increment over exposure	+log_tc	6,040	191	+0.042	—	—	—	—	DeLong $p = 2.1 \times 10^{-6}$
Score ranks excess contact	log_tc	16,843	475	0.764	0.095	0.150	0.158	5.32	$1[O_i - E_i > 0]$
Permutation (exposure-preserving)	log_tc	16,843	191	—	0.126	—	—	—	$p = 1.00$ (B)
Matched / stratified (CEM-opp)	log_tc	6,040	191	0.863	—	—	—	—	$\Delta = 0.083$

Notes: The validation target is adjudication-anchored exposure (cobidder status) among always-loser firms, not cartel membership. Expected contact $E_i = \sum_g n_{ig} p_g$ is built from procurement opportunity cells (item, buyer, year, modality); cell definitions and retained support are in [Appendix C.1](#). PR-AUC is the primary metric because the target is rare (191/16,843); ROC-AUC is reported alongside but is not the lead. The exposure-only benchmark reproduces the label at least as well as the raw score on both ROC-AUC and PR-AUC. The score’s isolated contribution is the nested DeLong increment (+0.042) and the within-opportunity-stratum AUC (0.771). Combined exposure-plus-score logits (ROC-AUC near 0.98) are deliberately omitted from this table because E_i mechanically encodes the label and their fit is not score evidence.

4.3. Timing, Leakage, and Case Composition

The previous subsection isolated a small, exposure-adjusted residual ordering. This subsection asks whether that ordering can be deployed, and the honest answer is a downgrade. The ranking is informative as a retrospective, incumbent-pool triage diagnostic, but it is concentrated in one adjudicated case and one procurement environment, and it does not support prospective, platform-wide deployment without first handling new entrants, score-zero ties, and cross-case breadth.

A retrospective rank measures; it does not deploy. Every benchmark anchors its labels to CADE adjudications that close years after the bidding they describe, so a high in-sample AUC certifies construct validity, not field performance. The strict-timing design freezes the score, the always-loser status, and the frequent-loser threshold to the 2009–2016 window and evaluates against 2017–2019 labels. There the signal survives *only* inside the always-loser training pool: the frequent-loser flag attains ROC-AUC 0.767 and the continuous score 0.750. On the realistic full candidate universe—every firm scorable at the screening date, with entrants carried at score zero—discrimination collapses to ROC-AUC 0.55, PR-AUC 0.005, and **precision@500 = 0 and recall@500**



Figure 2: Opportunity-adjusted diagnostic: observed defendant contact O_i versus expected contact E_i under random matching, by award-layer score bin. The positive gap $O_i - E_i$ in the top score bins is the within-opportunity signal; expected contact is built from procurement opportunity cells (item, buyer, year, modality). *Alt text:* A binned plot with award-layer score bins on the horizontal axis and defendant-contact counts on the vertical axis. Two series are shown per bin, observed contact and expected contact under random matching; observed contact exceeds expected contact in the highest score bins, indicating excess contact concentrated where the score is high.

$= 0$. A screen that returns zero true positives in its top 500 candidates is not deployable. The mechanical cause is structural: of the 193 positives, 45 (23%) are entrants with no pre-window history, so the frozen score cannot rank them, only 148 positives are rankable at all, and roughly a quarter of the universe is tied at score zero. A loser-side score built from past losing cannot order firms that have not yet lost. The rolling-origin design confirms this: on the full universe every test year from 2014 to 2019 returns PR-AUC near 0.004–0.006 and precision@500 and recall@500 of zero, with the worst year (2015) at ROC-AUC 0.50, below chance; the apparent within-pool improvement is retrospective accumulation of adjudicated labels, not forward prediction. The rolling-origin curves are in Appendix [Appendix C](#).

The decisive test is case composition. A single case (trens/metros, CADE process 08700.004617/2013-41) supplies 55% of all positives and 72% of the top-500 true positives. Dropping it collapses the operational metrics: PR-AUC falls from 0.126 to 0.036 (–71%), precision@500 from 0.130 to 0.040, and recall@500 from 0.342 to 0.233; dropping the top two cases leaves PR-AUC 0.029. ROC barely moves ($0.939 \rightarrow 0.929$), which is precisely why ROC is the wrong lead metric here—it is insensitive to the case concentration the operational metrics expose. Case-balanced precision@500 is 0.029 against the firm-weighted pooled 0.130; only six linkable cases contribute any top-500 true positives, the largest accounting for 64–87%. The same concentration appears across environments: dropping the largest item-group (31% of positives) roughly halves PR-AUC ($0.126 \rightarrow 0.062$), and 96.8% of positives sit in Pregão. A clustered randomization-inference test (buyer $\times$ item-group $\times$ year cells, $B = 1000$) confirms the ordering is real but thin: observed PR-AUC 0.126 beats the null at $p = 0.015$, yet case-coverage breadth is *not* significant ($p = 0.32$). The leave-one-case-out distribution is reported in Appendix [Appendix C](#); Table 4 synthesizes the full battery.

The most this design can support, even under strict timing, is a triage queue. A retrospective, incumbent-pool, single-case ordering tells an agency where to look first among firms it already observes; it cannot identify cartels *ex ante*, rank firms it has never seen, or be deployed platform-wide on the strength of one adjudicated cluster. The composite verdict is C (the operational signal is mainly retrospective and incumbent-pool)

and D (it is case-dominant and cluster-sensitive).

Table 4: Timing and Case-Composition Synthesis

Design	Information set	N	Pos.	ROC-AUC	PR-AUC	Prec@500	Rec@500	Leakage status	Interpretation
Full-sample retro-spective diagnostic	Full retrospective	16,843	191	0.939	0.126	0.130	0.342	Upper bound; all leaks	Construct validity, not timing
Relaxed temporal holdout	Score frozen; future wins used	16,843	193	0.864	—	—	—	Leaks future wins	Score-timing sensitivity only
Strict full test universe	Strict; entrants at score 0	41,444	193	0.55	0.005	0	0	No leak	Collapse: not deployable
Strict rankable (incumbent) sample	Strict; prior-loss firms	21,819	191	0.767	0.045	0.062	0.161	No leak	Survives only inside pool
Rolling-origin mean	Strict; full universe	—	—	0.56	0.005	0	0	No leak	Accumulation, not prediction
Rolling-origin worst year (2015)	Strict; full universe	29,804	127	0.50	0.004	0	0	No leak	Below chance
Leave-one-case-out mean	Full retrospective	16,750	—	0.95	—	0.029	0.522	Case-structured	Case-balanced precision tiny
Leave-largest-case-out	Full retrospective	16,750	86	0.929	0.036	0.040	0.233	Case-structured	—71% PR-AUC collapse
Leave-top-two-cases-out	Full retrospective	16,649	75	0.925	0.029	0.030	0.200	Case-structured	Thin residual tail
Clustered randomization inference	Label shuffle in cells	16,843	191	0.939	0.126	—	—	Permutation null	$p = 0.015$; coverage $p = 0.32$
Direct-defendant scope	Strict and full scores	—	—	0.49	—	—	—	No leak	Scope boundary, not a detector

Notes: PR-AUC, precision@500, and recall@500 are the primary metrics because the target is rare; ROC-AUC is reported alongside but is not the lead, and it is deliberately robust precisely because it is insensitive to the case concentration that the operational metrics expose. The full-sample retrospective row is a diagnostic *upper bound*, not a timing claim. In the strict and rolling-origin designs the score is frozen before the test window; later CADE adjudications are still used as labels but were not legally observable at the screening date, so these designs distinguish prospective scoring from retrospective adjudication. Cobidders are adjudication-anchored exposure labels, not cartel members. The strict full universe carries entrants at score zero; a quarter of that universe is tied at zero. “0” denotes an exact zero true positives in the top 500.

4.4. Direct-Defendant Scope and Bounded Reading

The direct-defendant exercise aligns the validation target with the screen’s legal-economic role. Direct CADE defendants are the legal anchors that locate adjudicated environments; they need not be persistent zero-win firms, since defendants may occupy winner-side, allocation, or rotating roles. The award-layer statistic ranks the loser side by construction, so the empirical prediction is asymmetry, and the data follow it: against direct defendants AUC is 0.49, close to random. A generic cartel-membership detector should rank defendants as well as cobidders; this one does not, which is the expected result for a loser-side ranking. A universe-anchored scope matrix sharpens the point—raw participation count on the full BEC universe against direct defendants returns AUC 0.383, *below* random, so a high-participation firm is less likely to be a direct defendant under the loser-side logic, reversing the direction rather than merely failing. This is consistent with the role separation in D4 (14.9% of direct defendants are always-losers; median win rate 0.261 versus 0.086 for cobidders). The full eight-row scope matrix is in Appendix [Appendix C](#).

Taken together, the benchmarks draw a boundary rather than certify a screen. Each addresses a distinct pre-registered threat—legal anchoring, participation volume, market exposure, timing, leakage, and scope asymmetry—and the consolidated threat map is in Appendix [Appendix C](#). Persistent zero-win participation orders adjudication-anchored loser-side exposure, and the ordering survives participation-volume and opportunity-set discipline as a significant residual. But that residual is modest, largely retrospective, concentrated among incumbents and in one adjudicated case, and silent on direct defendants. The defensible reading is therefore bounded: the award layer ranks where loser-side exposure is most likely worth bid-layer follow-up, within a reach the tests make explicit; it does not validate the legal status of ranked firms, nor support prospective, platform-wide deployment on its own. The next step is economic interpretation: among firms already selected by the frequent-loser rule, how far do cobidders display a distinct loser-side profile rather than a mechanical collection of high-volume losers concentrated in a few environments?

5. Economic Content and Ordinary-Loser Alternatives

The validation results show that the award-layer ranking concentrates adjudication-anchored loser-side exposure. This section gives that exposure target economic content through a demanding within-stratum comparison: among firms already inside the frequent-loser stratum, do the firms our screen targets—those observed co-bidding with direct CADE defendants—differ from other frequent losers in dimensions relevant for loser-side forensic prioritization? Throughout, “cobidder” denotes an adjudication-anchored exposure label, not legal membership. Because the samples are large (tens of thousands of firms), we report standardized mean differences (SMD, Cohen’s d) rather than p -values: significance is uninformative at this N , while standardized differences speak to economic magnitude. The full profile, opportunity-adjusted differences, monotonicity bins, binary-versus-continuous comparison, and the ordinary-loser adjudication table are reported in Appendix [Appendix D](#); we summarize the headline findings here.

5.1. *Economic content and network specificity*

Cobidders are economically distinct from other frequent losers, and not marginally so. The 191 cobidders are almost entirely electronic-auction firms—a Pregão share of 96.6% against 66.6% for the 2,544 other frequent losers (SMD 0.76)—they are deployed more broadly, reaching 24.9 distinct buyers versus 14.3 (SMD 0.72), they are more persistent across active years (SMD 0.84), and they sit far closer to the legal anchors used in validation (standardized difference in CADE cases touched 3.76). These are deployment and proximity measures, not legal conclusions: cobidders occupy a wider, longer-lived loser-side footprint within the already-selected frequent-loser pool, anchored to the adjudicated network. The ranking is also specific to that network. Negative controls that replace the true CADE defendants with matched placebo anchors and re-score yield a real-anchor ROC-AUC of 0.939 that dominates the placebo-anchor distribution (mean 0.76; empirical $p < 0.001$), and the global zero-win definition of always-losers is not dominated by any market-specific alternative (best alternative 0.774, well below the global 0.939). The screen’s ranking quality is tied to the actual defendant network, not to generic high-volume co-bidding—the central strength of the validation (Appendix [Appendix D](#)). This reading is consistent with evidence that auction collusion can operate through bidder roles, repeated relationships, and cartel organization ([Baldwin et al., 1997](#); [Porter and Zona, 1999](#); [Pesendorfer, 2000](#); [Asker, 2010](#)).

5.2. *An exposure ranking, not a collusion-intensity ranking*

Much of the raw distinction is procurement-environment composition and partly mechanical proximity that attenuates—and in one case vanishes—once we adjust for opportunity, the expected contact and exposure a firm would accumulate given its participation profile. The item-group concentration difference, raw SMD 0.45, *vanishes* under adjustment (adjusted SMD 0.02): the apparent product specialization of cobidders is procurement-environment composition, not an intrinsic portfolio trait. Proximity to the legal anchors roughly *halves*—the CADE-cases SMD falls from 3.76 to 2.25—consistent with a large part of raw proximity being the mechanical consequence of broader participation in defendant-bearing cells. Persistence survives; product specialization does

not (Appendix [Appendix D](#)).

The most informative diagnostic concerns what the ranking actually orders. Binning firms by participation T_i , cobidder prevalence rises monotonically with participation—from zero in the lowest bins to roughly 17% in the highest (rank correlation +0.92)—but opportunity-adjusted excess contact $X_i = O_i - E_i$ is flat-to-negative across the same bins (rank correlation -0.93). Higher-ranked firms are not contacting defendants *more than their exposure predicts*; they are simply more exposed. The award-layer rank is therefore an **exposure ranking, not a collusion-intensity ranking**—exactly what the reach-and-limits reframe expects. This also reframes the binary FL14 cutoff as an *administrative* implementation of the exposure ranking, not a structural break: the continuous score $\log(1+T_i)$ ties or beats the FL14 binary in every full-sample configuration (full-universe ROC-AUC 0.939 continuous vs. 0.924 binary), and there is no bunching at the threshold—the density ratio at $T = 14$ is 1.06, essentially flat (Appendix [Appendix D](#)). FL14 buys deployability and a simple reporting cut, not a behavioral discontinuity.

5.3. Ordinary-loser alternatives and what Section 5 establishes

Could the same profile arise from ordinary, non-collusive reasons that firms lose persistently? The honest summary is that ordinary-loser alternatives are *narrowed but not eliminated*. Cobidders are wider and longer-lived rather than narrow and short-lived, with a markedly lower Convite and minimum-quorum share, so they are not low-capacity quorum-fillers. But the proxies that would settle the remaining alternatives—distance-to-winner, geography and logistics, and later wins—are NOT_OBSERVED at the award layer, so benign specialization and geography remain possible (Appendix [Appendix D](#)). What distinguishes cobidders from these ordinary stories is the one axis the alternatives do not predict: contact with the adjudicated CADE network, the axis the screen is built on.

Section 5 therefore gives the award-layer exposure target genuine but bounded economic content. The strengths are real—cobidders are economically distinct and the ranking is specific to the real defendant network ($p < 0.001$ on ROC against matched placebo nulls)—but the limits are equally clear: much of the raw distinction is procurement-environment composition and partly mechanical proximity that attenuates under oppor-

tunity adjustment; the rank is monotone in exposure but flat in opportunity-adjusted excess contact; FL14 is administrative, not structural; and ordinary-loser alternatives are narrowed, not eliminated. The construct is a disciplined exposure ranking with genuine economic content. It does not establish membership, agreement, liability, or cover bidding in any tender, and it is not diagnostic by itself. Read together with the validation of Section 4 and the operational sequencing of Section 6, it supports triage, not liability: the award layer orders attention toward adjudication-anchored loser-side exposure, and richer records evaluate what the screen has prioritized. The next section turns from content to implementation, asking whether the award-layer rank can be placed before bid-distribution forensics in a sequence that saves bid-microdata recovery while preserving adjudication-anchored recall.

6. From Award-Layer Triage to Bid-Layer Forensics

The preceding sections validate the award-layer ranking on the margin for which it is designed. Section 4 shows that persistent zero-win participation orders adjudication-anchored loser-side exposure; Section 5 shows that the exposed firms have an economically distinctive profile rather than merely high procurement volume. The remaining question is how an enforcement agency should use such a signal. The answer is not to stop at the award layer, and it is not to replace bid-layer forensics. A thin administrative record can rank priorities, but it cannot by itself evaluate the bid conduct that richer cartel evidence requires.

This section turns from validation to sequencing. The screen is useful because it sits before costly proof-producing work: it uses records the agency observes broadly—who participated, who won, and who kept appearing without winning—whereas bid-layer forensics use submitted offers, within-tender bid moments, and distance from the winning bid, which are more probative but more expensive to recover, clean, and inspect at scale. The issue is not whether a cheap signal should replace a costly one, but whether it can discipline where the costly one is opened first. We evaluate that architecture in three steps: we place the award-layer ranking and an Imhof–Wallimann-style bid-distribution benchmark on the same adjudication-anchored target, so performance can be read against

information cost; we ask whether the two layers are complements; and we implement the operational gatekeeper, in which award records create a survivor pool and bid forensics rank firms inside it. The object is a cost-recall frontier for forensic attention, not a horse race under full observability. This is the paper’s central enforcement implication: administrative records rank where proof-producing forensics should begin, while bid-level evidence remains the stage at which conduct is evaluated.

6.1. Bid-Layer Forensics as a Full-Observability Benchmark

The two layers sit at different points in the evidentiary pipeline. The award layer uses records the agency already holds; the bid layer uses the submitted offers, which in the Brazilian setting are not in the routine award file and must be recovered from the underlying LANCES bid histories, harmonized, and reduced to within-tender moments before any firm can be scored. The bid-layer benchmark is therefore the forensic stage, not a free comparator: it becomes available only after the agency has paid the cost of opening the bid file. The right question is how much ranking information is already present before that file is opened.

The benchmark is a fair, transparent forensic comparator rather than a black box. The learner is a random forest (ranger, 500 trees, probability mode, no hyperparameter search), trained on seven within-tender Imhof–Wallimann moments—the coefficient of variation and its within-firm dispersion, skewness, kurtosis, spread, the log min–max ratio, and the second-lowest-bid gap—each averaged across the tenders a firm contests ([Abrantes-Metz et al., 2006](#); [Imhof et al., 2018](#); [Imhof, 2019](#); [Huber and Imhof, 2019](#); [Wallimann et al., 2023](#)). The candidate pool is the always-losers for which these moments are computable, the target is the same adjudication-anchored cobidder exposure used throughout Section 4, and all 193 positives are retained; the small attrition is mechanical and concentrated among non-positives. Timing and bid-revision features are excluded because the records carry no usable per-bid timestamps. The candidate-support audit, the model-design audit, and the leakage-boundary table are reported in full in Appendix [Appendix E](#).

Performance.. Table 5 reports precision, recall, and PR-AUC first and ROC second, since the positive base rate is near one percent and ranking quality is what an agency

cares about. Two facts stand out. First, in the pooled full-observability design the cheap award flag is *comparable* to the costly bid benchmark: the binary frequent-loser flag reaches ROC 0.921 and the bid-layer random forest reaches ROC 0.888, with overlapping confidence intervals—comparable discrimination at lower informational cost, not a victory of one layer over the other. The combined award-plus-bid model reaches ROC 0.962 and PR-AUC 0.275, but it requires bid microdata for every firm and is therefore a full-observability upper bound rather than a deployable screen. Second, the pooled metrics rest on *random* cross-validation folds that are optimistic because the cobidder positives cluster by CADE case. Under case-grouped folds, where the positives from a held-out case never co-train, the bid layer falls hardest: its PR-AUC drops from 0.124 to 0.045, a 63% reduction, and its ROC from 0.891 to 0.810. The combined model holds up (ROC 0.936) only because the label-independent award score continues to rank firms when the bid signal thins. Excluding the label-defining tenders lowers the bid ROC only modestly (to 0.874, -0.017), so the fragility is case clustering, not feature–target overlap.

Complementarity. The two layers carry partially different ranking information rather than duplicating one another: the Spearman rank correlation between the award and bid scores is only 0.42, and adding the award score to the bid benchmark improves PR-AUC by $+0.151$ under pooled folds (DeLong $p < 0.001$). We report the honest bound on that increment: under case-grouped folds it more than halves to $+0.064$ PR-AUC (still $p < 0.001$), and what survives is mostly the award score *rescuing* the bid layer rather than the reverse. The combined model is therefore an upper bound on what the two layers can jointly do under full observability, not an operational first-stage rule. Neither score is proof of an agreement: the bid benchmark measures the shape of a firm’s bid distribution, and false positives are non-labeled firms, not “bad firms.” A strict-timing version of the bid benchmark is *blocked*, because within-tender dispersion moments cannot be cleanly restricted to a pre-target window; the award score, by contrast, is label-independent and admits a clean temporal holdout. The full complementarity decomposition, rank-overlap diagram, and leakage boundary appear in Appendix [Appendix E](#). What the benchmark establishes is the discriminating ceiling under full observability, and how much of it the

cheap award layer already reaches at far lower informational cost.

Table 5: Bid-Layer Benchmark and Award-Layer Screen: Ranking Performance

Design	Model	PR-AUC	Prec@500	Rec@500	Prec@1000	ROC
Pooled (random folds)	Award continuous	0.126	0.130	0.342	0.095	0.938
	Award FL14	0.085	0.088	0.232	0.089	0.921
	Bid-layer random forest	0.124	0.136	0.358	0.100	0.888
	Combined award + bid	0.275	0.198	0.521	0.149	0.962
Case-grouped (held-out case)	Award continuous	0.126	0.130	0.342	0.095	0.938
	Award FL14	0.085	0.088	0.232	0.089	0.921
	Bid-layer random forest	0.045	0.054	0.142	0.059	0.810
	Combined award + bid	0.109	0.122	0.321	0.121	0.936
Excl. label-defining tenders	Bid-layer random forest	0.075	0.092	0.242	0.084	0.874

Notes: $N = 16,731$ always-losers with computable bid moments; 190 in-pool cobidder positives. PR-AUC is reported first because the positive base rate is near one percent. The **pooled** block is a full-observability diagnostic with random-fold optimism; the **case-grouped** block is the honest robustness test (positives clustered by case never co-train) and shows the bid layer falling hardest (PR-AUC $0.124 \rightarrow 0.045$, 63%). The award scores are label-independent and so do not change across designs. Cobidders are adjudication-anchored exposure, not cartel membership; false positives are non-labeled firms, not “bad firms.” The combined model is a full-observability upper bound, not a first-stage screen. Candidate support, model audit, complementarity, and the leakage boundary are in Appendix [Appendix E](#).

6.2. Sequential Gatekeeping and the Cost-Recall Frontier

Similar discrimination at lower cost does not by itself establish a useful two-stage architecture. The two layers sit at different points in the evidentiary pipeline, and that ordering is the operational lever. The award layer is routine and cheap: it uses records the agency already holds. The bid layer is richer and more probative but costlier: in this setting the submitted offers are not in the award file and must be recovered from the underlying LANCES histories, harmonized, and reduced to within-tender moments before any firm can be scored. A model that gives the classifier both layers for every firm—the *joint* model—is an upper bound on what the two layers can do under full observability, but it has already paid the full cost of opening the bid file. The implementable object is therefore sequential: rank where the costly layer should be opened first.

The sequential rule imposes that constraint directly. Firms are ranked using only the award layer; the top K_1 firms form a survivor pool S_{K_1} ; bid microdata are recovered *only* for the survivors in S_{K_1} ; and the bid-distribution benchmark then reranks firms within that smaller pool by the combined award-plus-bid score, producing the final top- k forensic queue. Full observability is the upper-bound benchmark, not the deployed policy;

award-only is the zero-bid-cost baseline that opens no bid records at all; the sequential rule is the operating architecture in between.

Cost is not only firms.. The first-order honest point is that the firm count is the wrong denominator for forensic burden. Recovering the within-tender bid moments requires the full tender-item LANCES record for every tender a survivor contested, so the relevant cost is measured in opened tender-items and opened bid rows, not in firms. Table 6 reports all three. At the $K_1 = 2,000$ operating point the rule opens only 12% of firms (a firm reduction of 88% relative to full observability), but it recovers 34% fewer tender-items and only 33% fewer bid rows. Survivors are the highest-participation always-losers, so opening 12% of firms still opens roughly 67% of the bid rows. The frontier reports firms, tender-items, and bid rows precisely so this divergence is visible: the honest processing-burden cut is about a third at the bid-row level, not the firm-level figure. We report the firm-vs-bid-row divergence (88% versus 33%) as the core caveat, not a footnote.

The frontier, not a cutoff.. Varying K_1 traces a cost-recall frontier rather than identifying a single best rule. At $k = 500$, sequential award→bid recall is non-monotone in K_1 : 0.34 at $K_1 = 500$, 0.44 at 1,000, 0.48 at 2,000, 0.52 at 3,000, and 0.42 at 5,000, peaking near $K_1 = 3,000$ (Fig. 3). At $K_1 = 3,000$ the sequential rule *matches* joint full-observability recall (0.52 versus 0.52 at $k = 500$) at roughly a 24% bid-row reduction, but it never exceeds the joint upper bound (0.52 at $k = 500$, 0.78 at $k = 1,000$). $K_1 = 2,000$ is *one operating point, not a calibrated optimum*: agencies with larger forensic capacity sit higher on the frontier and agencies with less capacity sit lower. The contribution is the frontier itself—the menu of recall-versus-scope trade-offs—not a universal cutoff.

What the bid rerank does, and does not, add.. The bid layer adds little operationally at a fixed survivor pool. At $K_1 = 2,000$ the award-defined survivor pool *already* contains 0.78 of the positives the final queue recovers; the subsequent bid rerank only reorders firms inside that pool and does not surface new positives. The architecture’s value is therefore in cost—preserving much of the target capture while reducing the scope of proof-producing forensics—rather than in the bid layer locating targets the award layer

missed. This is the opposite of a claim that the bid stage drives recovery: it is the award gate that determines which positives are reachable, and the rerank that orders them.

False positives and missed positives.. The frontier is explicit about both error margins. At $K_1 = 2,000$, $k = 500$, the rule opens 1,852 non-cobidder firms for bid recovery, loses 42 positives at the award gate (they never enter the survivor pool), and ends with 409 non-labeled firms in the final top-500 queue; 99 of the 190 positives are missed overall, and precision is modest at about 0.18. We do not read modest precision as failure: in forensic triage, high recall that routes a manageable queue to the next, costlier evidentiary stage is the operative property, and the 409 non-labeled firms in the queue are unverified, not exonerated.

Baselines.. The frontier is bracketed by three reference points. Award-only is the *zero-bid-cost* baseline (recall 0.34 at $k = 500$), achievable with no bid recovery whatever. Bid-only and joint scoring are full-observability benchmarks that require opening the entire pool's bid records; joint is the upper bound the sequential rule approaches but never exceeds. A random survivor pool of the same size is far below the award-ranked one (recall 0.10 at $K_1 = 2,000$, three to twelve times worse across operating points), confirming that it is the award ranking, not the mere act of opening K_1 firms, that carries the signal.

The institutional reading follows directly. The rule is not designed to beat full information; it preserves much of the target capture while reducing the proof-producing forensic scope—and even that reduction is honest only at the firm level and modest at the bid-row level. The bid rerank's marginal contribution at a fixed survivor pool is reordering, not new recall. A screen of this kind does not decide liability; it decides, under a stated capacity constraint, which files deserve the next and more expensive form of evidentiary attention.

The frontier is also case-fragile. Leaving out the single largest CADE case—which supplies 55% of the positives—drops sequential recall at $K_1 = 2,000$ from 0.48 to 0.34 at $k = 500$. The signal is real but concentrated; a different case mix would move the frontier, and the operating points should be read as setting-specific rather than portable magnitudes.

Table 6: Cost-Recall Frontier for Sequential Award→Bid Gatekeeping

Rule	K_1	Firms open	Tender items	Bid rows	TP	FP	Recall	Prec.	Δ Firm red.	Δ Bid red.	RLoss vs joint
<i>Panel A: forensic queue $k = 500$</i>											
Award-only (no bid)	—	500	41,704	1,356,346	65	435	0.34	0.13	0.97	0.60	0.18
Bid-only (full obs)	—	16,772	116,112	3,393,915	66	434	0.35	0.13	—	—	0.17
Joint (full obs, U.B.)	—	16,772	116,112	3,393,915	99	401	0.52	0.20	—	—	0.00
Sequential	500	500	41,704	1,356,346	65	435	0.34	0.13	0.97	0.60	0.18
Sequential	1,000	1,000	58,408	1,783,750	83	417	0.44	0.17	0.94	0.47	0.08
Sequential	2,000	2,000	77,097	2,283,924	91	409	0.48	0.18	88%	33%	0.04
Sequential	3,000	3,000	87,714	2,575,274	99	401	0.52	0.20	0.82	0.24	0.00
Sequential	5,000	5,000	99,977	2,921,977	79	421	0.42	0.16	0.70	0.14	0.11
<i>Panel B: forensic queue $k = 1,000$</i>											
Award-only (no bid)	—	1,000	58,408	1,783,750	95	905	0.50	0.10	0.94	0.47	0.28
Bid-only (full obs)	—	16,772	116,112	3,393,915	99	901	0.52	0.10	—	—	0.26
Joint (full obs, U.B.)	—	16,772	116,112	3,393,915	149	851	0.78	0.15	—	—	0.00
Sequential	1,000	1,000	58,408	1,783,750	95	905	0.50	0.10	0.94	0.47	0.28
Sequential	2,000	2,000	77,097	2,283,924	127	873	0.67	0.13	88%	33%	0.12
Sequential	3,000	3,000	87,714	2,575,274	140	860	0.74	0.14	0.82	0.24	0.05
Sequential	5,000	5,000	99,977	2,921,977	113	887	0.59	0.11	0.70	0.14	0.19

Notes: Evaluation pool: 16,772 always-loser firms with computable complete-Imhof bid-distribution features; positives are 190 CADE-cobidders. The target is *adjudication-anchored exposure*, not cartel membership. “Firms open,” “Tender items,” and “Bid rows” report the forensic footprint that must be recovered before the rule’s final queue can be scored; full observability opens all 16,772 firms / 116,112 tender-items / 3,393,915 bid rows. **The firm count is not the only denominator:** opened tender-items and bid rows better approximate processing burden, and the firm reduction (Δ Firm red.) substantially overstates the bid-row reduction (Δ Bid red.) because survivors are the highest-participation firms. **Award-only opens no bid records** as a pure award screen; the bid-footprint columns for the award-only rows are the *hypothetical* scope if it were verified at the bid layer. Bid-only and joint require full bid observability for the entire pool; the sequential rule recovers bid microdata only for the K_1 survivors. Cost reductions are computed relative to full observability. “RLoss vs joint” is recall lost relative to the joint upper bound at the same k . False positives (FP) are non-labeled firms, *not* exonerated. ROC is not the decision metric here; recall against the adjudication-anchored target at a fixed forensic queue size is.

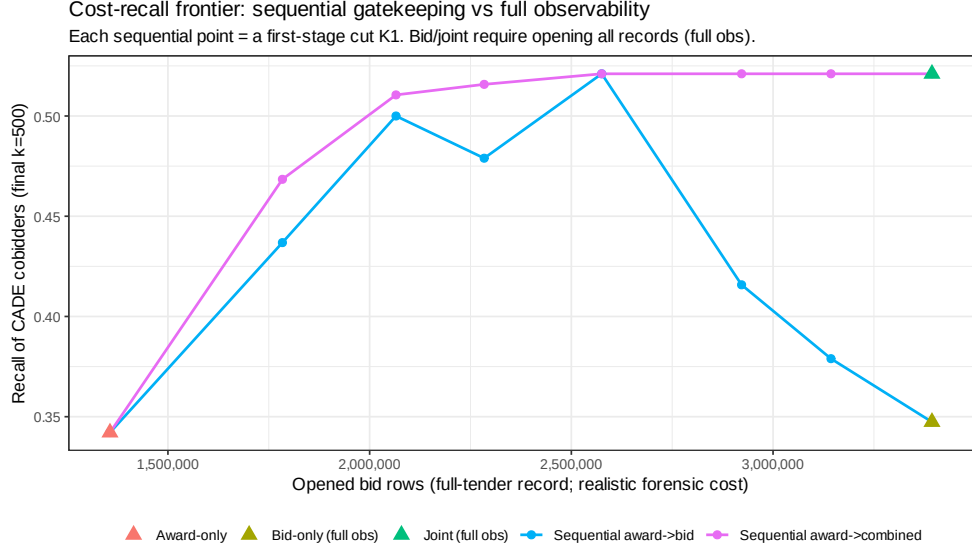


Figure 3: Cost-recall frontier for sequential award→bid gatekeeping. Points trace operational trade-offs in forensic attention, *not* legal proof. The horizontal axis is the recovered bid-record footprint (bid-row reduction relative to full observability) and the vertical axis is recall against adjudication-anchored cobidder exposure at a fixed forensic queue size. Full-observability benchmarks (bid-only, joint) require whole-pool bid recovery and are plotted as horizontal reference lines; the sequential points vary $K_1 \in \{500, 1,000, 2,000, 3,000, 5,000\}$ and never exceed the joint upper bound; award-only is the zero-bid-cost baseline.

Alt text: A scatter-and-line plot. The horizontal axis is the reduction in recovered bid records relative to opening every firm’s bid history; the vertical axis is recall of adjudication-anchored cobidder positives at a fixed forensic queue size. A curve of sequential operating points rises and then falls as K_1 grows from 500 to 5,000, peaking near $K_1 = 3,000$ where it touches a horizontal dashed line marking the joint full-observability upper bound; it never rises above that line. A second horizontal line marks the lower bid-only full-observability benchmark, and a marker at the far right shows the zero-bid-cost award-only baseline at lower recall. The visual message is that opening fewer bid records costs recall along a frontier, the sequential rule at best matches but never exceeds full information, and the firm-level cost saving overstates the bid-row cost saving.

Timing discipline. The frontier above uses the full feature window, which is close to the adjudication target. A stricter timing audit restricts the award-layer score to a 2009–2016 window and evaluates against later adjudication-anchored exposure; this is not a real-time field deployment, since the labels remain adjudication-based, but it disciplines the information used to rank firms before bid forensics are allocated. Under that restriction the *award-only* score remains usable, with ROC falling to 0.73 from the full-window 0.94. The *sequential* rule, by contrast, cannot be cleanly time-limited: the within-tender bid moments that drive the rerank cannot be restricted to a pre-target window without contaminating them with post-window tenders, so a strict-timing sequential frontier is *blocked*. We state this rather than fabricate a number. The conclusion of Section 6 is therefore operational and bounded: cheap administrative records can structure the order in which expensive proof-producing evidence is assembled and, on the award layer, survive a timing holdout; the richer bid layer remains the place where conduct is evaluated, and its strict timing properties cannot be certified here.

7. Scope, Limits, and Price Corroboration

Section 6 carries the paper’s central enforcement-design result. Price evidence enters here in a narrower, corroborative role only. It is scope evidence: it speaks to whether the award-layer screen points to economically non-neutral environments. It is *not* a damages estimate, an overcharge, a cartel markup, a causal price effect of frequent-loser presence, or proof of a cover-bidding mechanism. Overcharge measurement has its own evidentiary tradition (Bryant and Eckard, 1991; Connor, 2007), distinct from screening (Marshall and Marx, 2012; Harrington, 2008). We state this boundary once and hold to it throughout. The full price regressions are reported in Appendix [Appendix F](#).

The broad price imprint. In the broad item-level regressions, frequent-loser presence is positively associated with log negotiated unit price: 0.0677*** with item and year fixed effects and 0.0636*** after adding procuring-unit fixed effects, with an implied range of +3.6%– +7.7%. The association is larger in pregão (0.0933***) than in convite (0.0382**). If frequent losers were only mechanical quorum fillers under the three-bidder rule, the

imprint should be stronger in convite; the larger association in the electronic-auction environment weighs against a pure procedural-artifact reading. These are descriptive associations used to assess economic non-neutrality, not causal effects.

The sign reversal, decomposed.. The positive broad-sample coefficient ($++0.064$, $p = 0.003$) reverses under overlap-cell ATT weighting to -0.097 ($p < 0.001$); under propensity-score trimming it becomes -0.307 . Only 1.06% of treated items lack a within-cell counterfactual, so the reversal is a weighting result, not a dropped-observation artifact. The two coefficients measure different objects. The broad sample mixes a positive *selection* component—cartels concentrate in cells with structurally higher underlying prices: among non-treated items, mean log price rises from 1.35 (Q1) to 6.93 (Q5) across cell FL-share quintiles, a $\Delta = 5.58$ log-point gap, with a partialled-out coefficient of $+3.55$ (SE 0.23, $N = 1,439,255$)—with a negative *within-cell* component, the overlap-cell ATT, which isolates the latter. The progression is monotone: broad-sample $++0.064$; overlap-cell unweighted $++0.044$ ($p = 0.035$); overlap-cell ATT -0.097 ($p < 10^{-9}$). The negative ATT reading survives both modalities (Convite -0.099 , Pregão -0.098), every PBU-size quartile, and tender-value quartiles Q1–Q3; the only systematic positive cell is the high-value tail ($+0.041$, $p = 0.045$), where coordination costs are spread over larger contracts. The direct-CADE overlap estimate is null (-0.061 , $p = 0.45$): price regressions do not recover the legal object of the case and supply no damages base. Full regressions and the bidder-count boundary appear in [Appendix F](#).

An honest limit on the mechanism.. The within-cell price compression loads on the count of genuine bidders, not on the frequent-loser count: adding $\log(n_firms)$ drives the within-cell coefficient from -0.048 to $+0.008$ (not significant). The within-cell price movement therefore operates through the bidder-count channel, and we do not claim to have identified a cover-bidding mechanism from the price record; the mechanism is *not* identified by these data. The decomposition establishes only that the broad positive imprint is selection-into-high-price-cells rather than a treatment effect, and that price evidence stops aligning with the loser-side scope outside the high-value tail. The institutional implication stands: the award-layer signal is a trigger for richer evidence

collection, not a stand-alone measure of price injury. Price evidence may be consistent with economic relevance, but it does not transform an award-layer statistic into proof of agreement or price injury. Award records rank where costly bid-level and documentary evidence should be assembled; bid records, case documents, and legal process determine whether the conduct supports an antitrust case. The architecture is more portable than the estimates: another agency should validate the ranking against its own enforcement anchors and observability constraints rather than importing the BEC cutoff or performance metrics mechanically.

7.1. Adaptive Deployment

The screen should be deployed as a moving queue rather than a static legal trigger. Any administrative rule can generate strategic response once firms learn which observable patterns attract attention; firms could rotate loser roles, engineer occasional wins, fragment identities, or avoid a known participation threshold. That possibility does not undermine the sequencing architecture, but it rules out naive use of the binary cutoff as a permanent enforcement boundary—a standard concern for cartel enforcement and screening (Green and Porter, 1984; Haltiwanger and Harrington, 1991; Harrington, 2008; Sánchez Graells, 2019). Agencies should use continuous ranks or capacity-constrained top- k queues, refresh thresholds, monitor bunching below operational cutoffs, and combine award-layer screening with bid-layer forensics whenever bid files can be recovered. The survivor-pool size of the refreshed queue is itself a point on the cost–recall frontier of Section 6, so deployment capacity and gaming-robustness are chosen together rather than tuned in isolation. A firm that moves to avoid one observable pattern may create another: discontinuous participation, suspicious identity changes, or shifts in bid-level behavior. This is also the right legal posture. The screen is not a rule of liability and should not be used as one. It is a way to order scarce investigative attention when proof is costly and the first observable record is legally thin—a disciplined place to begin without pretending that the beginning is the end of the evidentiary process.

8. Conclusion

Cartel enforcement often begins before the agency has the evidence needed to prove a case. Procurement records then pose a problem prior to liability: how should scarce investigative capacity be allocated when the cheap administrative record is broad but legally thin, and the richer bid-level record is costly to recover and evaluate? This paper treats procurement-cartel screening as evidence allocation under costly proof.

The empirical design separates the award layer from the bid layer. The award layer observes participation and outcomes before bid histories are opened. The bid layer observes submitted offers and within-tender conduct; it remains the natural stage for proof-producing forensic evaluation. The contribution is not to collapse these layers, but to sequence them: administrative records rank where investigation should begin, and bid-level evidence evaluates conduct inside the prioritized set.

The validation evidence supports that division of labor within clear limits, and establishing those limits is the point. Persistent zero-win participation orders adjudication-anchored loser-side exposure, and the ordering carries a statistically significant residual once participation volume and opportunity-set exposure are held fixed. We do not stop there. A disciplined decomposition—opportunity-cell adjustment, strict rolling-origin timing, leave-one-case-out, and environment-dominance checks—shows that the residual is modest and bounded: most of the raw discriminating power is procurement opportunity; the operational performance is concentrated in a single adjudicated case and one procurement environment; and the ranking discriminates retrospectively among incumbent firms rather than prospectively across the platform, where new entrants cannot be ranked by prior losing. The ranking does not behave like a generic direct-defendant detector, consistent with its loser-side scope. Separating this genuine, bounded signal from exposure arithmetic and case concentration—and saying plainly where each begins—is the paper’s methodological contribution.

A complementary and more affirmative result concerns the two layers’ division of labor. On the same adjudication-anchored target, the award-layer score carries information beyond a bid-distribution benchmark, so award and bid signals are complements rather

than substitutes. Used as a gatekeeper within its boundary conditions, the award-layer ranking concentrates the bid-microdata pool the forensic stage must open while recovering a large share of adjudicated cobidders. Joint scoring is a full-observability upper bound; sequential gatekeeping is the architecture relevant once an agency knows where the cheap layer ranks and where it does not.

The limits are part of the design. The screen creates forensic priority, not proof. The validation target is adjudication-anchored exposure, not cartel membership. Price evidence is scope evidence, not damages. The architecture is portable only where award records, repeated participation, comparable procurement categories, and recoverable bid files coexist. A static cutoff can be gamed, so implementation should use refreshed thresholds, continuous ranks or top- k queues, bunching checks, and bid-layer follow-up.

Cheap suspicion should neither be ignored nor treated as proof. Award-layer triage gives an agency a disciplined way to order scarce investigative attention before the richer evidentiary record is assembled—and, just as usefully, a disciplined way to know when a cheap administrative signal is mostly exposure and concentration rather than evidence. The statistic indicates where legal attention should start, not where legal responsibility ends; mapping that reach, and its edge, is what makes the cheap layer safe to use.

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